K-Means Clustering Algorithm

The 5 Fundamental Steps

Step 1: Determine the Number of Clusters (K)

The first step is to decide how many clusters (groups) you want to categorize your data into. This number is denoted as 'K'. Techniques like the "Elbow Method" are often used to find the optimal value for K by plotting the Within-Cluster Sum of Squares (WCSS).

Step 2: Initialize Centroids Randomly

Once K is decided, the algorithm randomly selects K data points from the dataset to act as the initial "centroids" (the center points of the clusters).

Step 3: Assign Points to the Nearest Centroid

For every single data point in the dataset, the algorithm calculates its distance (usually Euclidean distance) to all K centroids. Each data point is then assigned to the cluster of the centroid that it is closest to.

Step 4: Update Centroids

After all points have been assigned to clusters, the algorithm recalculates the centroids. The new centroid for a cluster is found by calculating the mean (average) of all the data points currently assigned to that cluster.

Step 5: Check for Convergence (Repeat if necessary)

The algorithm compares the new centroid positions with their previous positions. If the centroids have moved, the algorithm loops back to Step 3 and reassigns points based on the new centroids. This process repeats until the centroids no longer move (or a maximum number of iterations is reached), meaning the clusters have stabilized.

Source Note: This document was extracted based on the geometric intuition and clustering explanation from the CampusX video "K-Means Clustering Algorithm | Geometric Intuition | Clustering | Unsupervised Learning".